DriftSense: Privacy-Preserving Intention-Aware Prediction and Model-Assisted Reflection for Browser-Based Digital Drift

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Digital wellbeing tools commonly rely on domain blocking, screen-time limits, and usage dashboards. These approaches can misclassify purposeful browsing because the same website can support focused information seeking, an intentional break, or unplanned consumption. We present DriftSense, a privacy-preserving Chrome extension and local machine-learning pipeline for studying browser-based digital drift, defined as a participant-reported mismatch between a declared browsing intention and the session outcome. DriftSense records only configured-domain metadata, aggregate interaction counts, and short activity-window summaries; it does not collect page content, screenshots, messages, or keystroke values. We specify a two-phase field protocol with approximately 20 adult Chrome users. During a 10-day collection phase, participants declare an intention and provide a post-session reflection while no model-assisted prompt is shown. Time, domain, intention-only, activity-only, and intention-plus-activity models are then evaluated, and a three-minute model and intervention threshold are frozen. During a seven-day intervention phase, elevated-risk sessions are randomized with equal probability to a reflective prompt or a silent control, while both assignments retain the same intention and post-session questions. Prediction is evaluated with participant-aware and chronological validation, and the intervention is evaluated by comparing self-reported drift in randomized eligible sessions. Empirical collection has not yet occurred, so this manuscript reports the system and methodology without invented results. DriftSense studies self-reported intention alignment; it does not claim to detect attention, emotion, addiction, or mental health status.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Ubiquitous and mobile computing systems and tools*; • **Computing methodologies** → *Machine learning approaches.*

Additional Key Words and Phrases: digital wellbeing, browser extension, intention alignment, digital drift, self-report, privacy-preserving sensing, session-level prediction, reflective check-ins

1 Introduction

People often open distracting websites for legitimate reasons. A user may visit YouTube to find a tutorial, Reddit to answer a technical question, LinkedIn to check a message, or a news site to follow a specific event. The same sites can also support unplanned passive consumption or avoidance. This ambiguity creates a problem for common digital wellbeing tools: a domain category or elapsed time alone cannot reliably distinguish purposeful use from digital drift.

Existing tools often operationalize undesirable use through screen time, app or website category, or fixed blocking rules [3, 13–15]. Such mechanisms can be useful for self-control, but they can also generate false positives when a user has a valid reason to visit a monitored website. They may also generate false negatives when a short session is clearly misaligned with intention. Recent work has therefore shifted toward user agency, reflection, friction, and intention alignment. For example, self-nudging systems such as *one sec* introduce a short pause before app access [9]; Purpose Mode reduces distracting interface patterns while preserving access to social media websites [11]; and PauseNow frames digital wellbeing as alignment between intentions and behavior [17].

We use the term *browser-based digital drift* to describe a session-level intention-behavior mismatch: the user begins with one declared purpose but later reports that the visit did not match that intention. This framing is deliberately narrower than addiction, attention detection, or productivity optimization. DriftSense does not infer clinical status or inner mental state. Instead, it collects explicit intention and explicit post-session reflection, then evaluates whether those self-reported labels can be predicted from lightweight browser-session signals.

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DriftSense is designed around a local-first Chrome Manifest V3 extension and a reproducible machine-learning pipeline. Before accessing a configured domain, the user declares an intended activity using a neutral set of options: *work or study task*, *learning or tutorial*, *specific information*, *communication or community*, *planned entertainment or break*, *open-ended browsing*, or *opened accidentally*. The categories avoid assuming that work is inherently aligned or leisure is inherently drift. During the session, the extension records duration, click count, scroll count, idle time, focus changes, and short activity-window summaries. After the session, it asks whether the visit matched the original intention. In the intervention phase, a locally computed three-minute risk estimate determines eligibility for a session-level randomization: half of elevated-risk sessions receive a brief reflective prompt and half form a silent control. The website remains accessible in either assignment.

The central research questions are:

> *RQ1: Can declared browsing intention combined with lightweight browser activity signals predict self-reported session-level drift better than time-based or domain-based approaches?*
> *RQ2: Among sessions classified as elevated risk after three minutes, does a model-assisted reflective prompt reduce subsequent self-reported drift compared with a randomized silent control?*

The expected contribution is an implementation and evaluation framework for intention-aware drift prediction and lightweight reflective support in browser sessions. The contribution is not activity tracking, dashboarding, or blocking alone. It is the combination of privacy-preserving activity signals, intention-grounded self-report labels, explicit comparisons against weak baselines, early prediction without future-session leakage, and a session-level randomized test of model-assisted reflection. The 10-day collection phase is used for model development; it is not treated as a causal control for the later intervention phase.

2 Related Work

2.1 Digital Self-Control Interventions

Digital self-control tools commonly use blocking, time limits, monitoring, reminders, friction, or interface modification [3, 13, 14]. Reviews find promising short-term effects but also short deployments, nonstandard outcomes, homogeneous samples, and few control groups; time spent is common but too coarse to represent why a site was used [14, 15]. Purpose Mode instead lets users suppress attention-capturing interface patterns while preserving access and found that goal alignment was more informative than site context alone [11]. These findings justify retaining time and domain as baselines rather than treating them as ground truth.

2.2 Intention and Reflection

Several systems create a reflective moment around technology use. *one sec* adds pre-access friction and an explicit choice to continue [9]; PauseNow proposes intention alignment but reports no outcome evaluation [17]; and StayFocused and MindShift deliver reflective or generated support, with mixed behavioral effects and concerns about repetitive prompts, assumptions, and richer personal data [12, 16]. WellScreen links estimated and actual smartphone use to daily reflection, but its small two-week deployment had no control condition [2]. DriftSense uses a shorter, nonblocking check-in and does not collect mental state.

Table 1. Closest systems informing the DriftSense research gap.

Work	Timing and mechanism	Role of intention	Evaluation	Remaining gap
Purpose Mode [11]	During-use interface modification	Goal alignment measured in EMA	Two-week field deployment	No session prediction; fixed phase order
one sec [9]	Friction before app access	Opening reason elicited around nudge	Observational field study and online experiment	No within-session prediction or field control
PauseNow [17]	Proposed recognition and nudging	Original intention is central	Conceptual framework	No deployed outcome evaluation
Before You Scroll Again [1]	Prediction before social-media use	Mismatch reported after use	Seven-day sensing study; within-person and held-out tests	Regret label; no intervention test
WellScreen [2]	Daily post-use reflection	Estimated versus actual use	Two-week mixed-methods deployment	No session classifier or control
DriftSense (planned)	Three-minute model-assisted reflection	Declared before and revisited after use	Ten-day collection plus randomized seven-day intervention	Browser-only pilot with short follow-up

2.3 Session-Level Modeling and Research Gap

Regret depends on activity and intention, not only duration [7, 10]. In a seven-day study with 21 participants, perceived intention–use mismatch correlated more strongly with regret than duration; phone-context CatBoost reached AUC 0.732 within person but only 0.536 when a participant was held out [1]. This gap motivates separate known-user and unseen-user evaluation. Behavioral sensing also creates privacy risks, so DriftSense limits collection to hostname-level context and aggregate interaction signals and makes no clinical or mental-state inference [4, 5].

Table 1 shows the closest comparison points. No evaluated system combines a declared pre-session browser intention, content-free early activity, a post-session alignment label, explicit weak baselines, and randomized model-triggered reflection.

The contribution is therefore narrow: evaluate whether intention plus lightweight activity improves session-level prediction over time, domain, intention-only, and activity-only alternatives, then test the resulting prompt against a randomized silent control. The short pilot cannot establish durable behavior change or population-level generalization.

3 System Design

3.1 Evidence-Derived Design Requirements

DriftSense is designed for a research setting where participants voluntarily install a browser extension, configure monitored domains, and export a local dataset for analysis. Five requirements follow from the reviewed evidence:

(1) **Measure alignment, not assumed harmfulness.** Screen time and domain are retained as baselines, but the target label must come from the participant's declared purpose and later reflection. This follows evidence that goal alignment and intention–use mismatch can be more informative than duration alone [1, 11].

(2) **Intervene lightly and preserve agency.** Reflection should provide an explicit choice and never block the site. A single neutral check-in follows evidence that friction and an opt-out can disrupt automatic action, whereas repetitive or inflexible prompts can be irritating [9, 12].

(3) **Minimize and localize sensing.** The system should use aggregate interaction counts and hostname-level context while excluding content, screenshots, messages, and key values. Local storage, inspection, export, and deletion must be participant controlled.
(4) **Keep measurement stable across conditions.** The same intention and post-session questions should be used throughout collection and intervention so that the mid-session prompt is the only randomized interface difference.
(5) **Make prediction and intervention auditable.** Exported records should support weak baselines, one-, three-, and five-minute prediction, participant-aware validation, a frozen deployment model, and verification of every eligibility and randomization decision.

Figure 1 summarizes the planned DriftSense extension and local machine-learning pipeline.

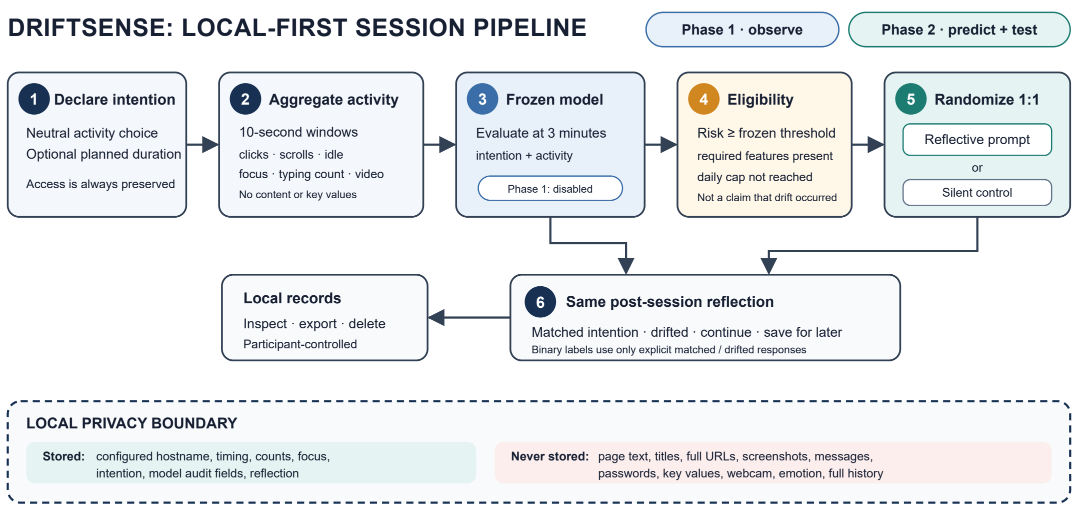


Fig. 1. DriftSense architecture and intervention decision flow. The prediction and randomization steps are disabled during the 10-day collection phase and enabled with a frozen model during the seven-day intervention phase.

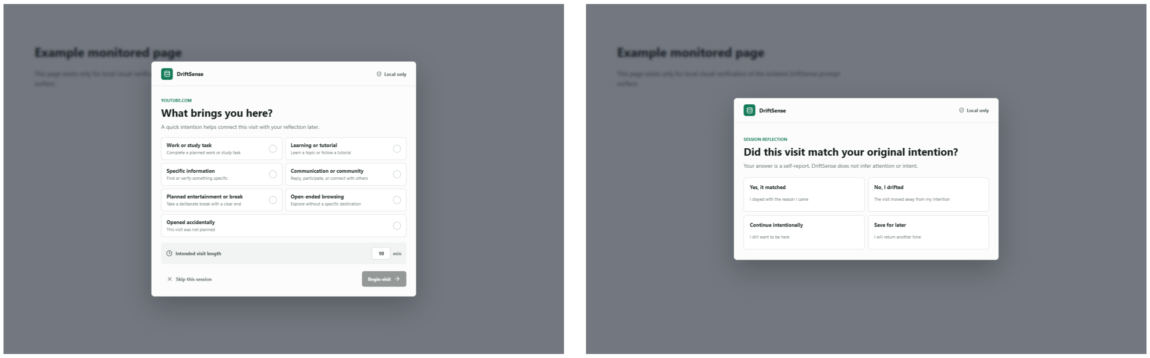
3.2 Chrome Extension Architecture

The collection prototype uses Chrome Manifest V3 with React and TypeScript. It contains a background service worker for session monitoring, content scripts for aggregate activity counting on monitored domains, a popup for quick status, an options page for domain configuration and privacy controls, and a dashboard for summaries and export. The intervention build will add a fixed local model evaluated after three minutes, an elevated-risk threshold, a 1:1 session-level randomizer, and a reflective prompt. No remote inference service is required.

Figure 2 shows the two participant-facing dialogs already implemented for the collection phase.

3.3 Session Flow, Sensing, and Labeling

Users select domains they consider drift-prone or worth studying; all other browsing is ignored. On opening a configured domain, they answer *Why are you opening this?* using one of seven neutral categories: work/study, learning/tutorial, specific information, communication/community, planned entertainment/break, open-ended browsing, or accidental opening. These categories do not assume that work is aligned or leisure is drift.



(a) Intention declaration (b) Post-session reflection

Fig. 2. Implemented participant-facing DriftSense interface for the collection phase. The dialogs are rendered by the actual extension component over a local example page: (a) captures intention and intended duration before monitoring begins, and (b) records the participant's session outcome. The planned Phase 2 model-assisted mid-session prompt is not depicted as implemented.

A session starts when a configured domain is opened or focused and ends when the tab closes, leaves that domain, times out, or completes reflection. The session export contains anonymous participant and session identifiers, start time, hostname, declared intention and intended duration, total duration, click/scroll/keyboard-activity counts, idle time, focus losses, and the reflection label. Every 10 seconds, a linked activity record stores only offset, event counts, idle/focus state, and accessible video-playback state. A separate intervention log records the frozen model version, threshold decision, eligibility, assignment, prompt delivery/response, and suppression reason. Exact field names remain in the exported-schema documentation.

After each session, both study phases and both randomized assignments use the same question:

Did this visit match your original intention?

The primary binary labels are defined only by explicit *Yes* and *No, I drifted* responses:

$$y = \begin{cases} 0, & \text{if answer is } \textit{Yes} \\ 1, & \text{if answer is } \textit{No, I drifted} \end{cases} \tag{1}$$

Continue intentionally and *Save for later* are retained as non-binary outcomes and excluded from the primary binary classification and intervention analyses. Missing answers also remain unlabeled; neither non-binary nor missing outcomes are recoded as non-drift.

During Phase 2, elevated-risk sessions assigned to the prompt arm receive *Still here for your original reason?* with *Continue intentionally* and *Finish now*; silent-control sessions receive no mid-session interface. The probability is not displayed, and the website remains accessible. The local dashboard supports aggregate inspection, pausing, domain configuration, export, and deletion. It is supporting infrastructure rather than a claimed contribution.

4 Drift Prediction and Model-Assisted Reflection

4.1 Prediction Task

The main task is binary session-level prediction:

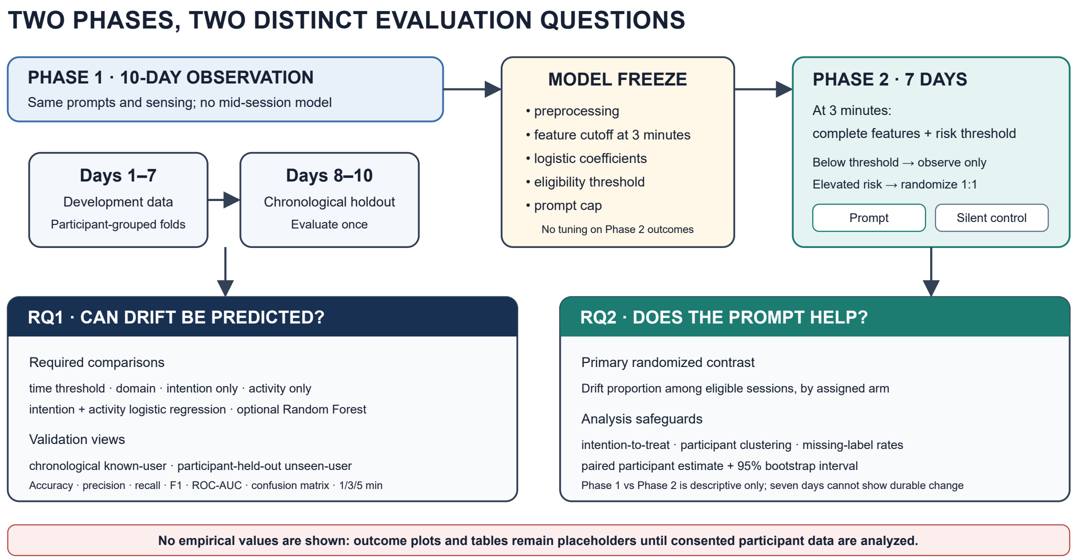


Fig. 3. Planned study and evaluation pipeline. RQ1 evaluates prediction using Phase 1 only; RQ2 evaluates the prompt using randomized eligible Phase 2 sessions. The Phase 1–Phase 2 change is not the causal intervention contrast.

> Given declared intention, domain metadata, and lightweight activity signals from a prompted browser session, predict whether the user will later report that the session was drift.

The positive class is the self-reported *No, I drifted* label. The negative class is the self-reported *Yes* label. Sessions labeled *Continue intentionally* or *Save for later* are analyzed separately or excluded from the binary task. The resulting model outputs a session-level drift probability. During Phase 2, crossing the frozen threshold determines eligibility for prompt-versus-silent-control randomization; the probability is not treated as attention detection.

4.2 Feature Groups

Features cover declared intention; domain category; time and day; duration so far; click, scroll, keyboard-activity, focus-loss, and tab-switch counts; idle, active, and video ratios; and activity rates. Activity windows are truncated at one, three, and five minutes, with the three-minute set used for deployment and the other cutoffs plus full session used offline. No early feature uses final duration or later activity.

Figure 3 shows the planned preprocessing, modeling, and evaluation pipeline for tabular drift prediction and model-assisted reflection.

4.3 Baselines

DriftSense evaluates four baselines before the main models:

(1) **Time-threshold baseline.** Predict drift if session duration exceeds a threshold tuned on the validation set.
(2) **Domain baseline.** Estimate each domain's drift tendency from training data only and classify validation or test sessions using the frozen training-derived rule.
(3) **Intention-only logistic regression.** Predict from declared intention and basic context features only.

(4) **Activity-only logistic regression.** Predict from aggregate activity signals without declared intention.

These baselines test whether the proposed intention-plus-activity models add value beyond common digital wellbeing heuristics.

4.4 Main Models

The primary model is intention-plus-activity logistic regression. It is selected for interpretability, modest data requirements, and straightforward local deployment as a fixed preprocessing specification, coefficient vector, intercept, and threshold. A Random Forest provides an optional nonlinear robustness comparison if the number and class balance of labeled sessions are adequate. CatBoost or XGBoost may be reported only as secondary offline models [6, 8]; they are not required for intervention deployment. Deep sequence models are outside the first study because 15–20 participants are unlikely to provide enough independent data to justify them.

4.5 Early Prediction

Offline evaluation compares one-, three-, and five-minute cutoffs with the full session. The three-minute model is chosen in advance for intervention deployment because it provides an early but non-immediate reflection point. Model family, preprocessing, threshold, and prompt cap are frozen before Phase 2. Exceeding the threshold makes a session eligible for local randomization; it does not automatically show a prompt or assert that drift occurred.

5 Methodology

5.1 Study Design

We plan a 17-day, two-phase, in-the-wild pilot with one cohort. Phase 1 is a 10-day observational collection period with consistent pre-session intention and post-session reflection prompts but no model-assisted mid-session intervention. After a short model-development interval, Phase 2 is a seven-day intervention period using the frozen model. In Phase 2, sessions that remain active for three minutes and exceed the frozen risk threshold are randomized 1:1 to a reflective prompt or a silent control. This micro-randomized comparison is the primary basis for RQ2. A simple before–after comparison between Phase 1 and Phase 2 will be reported only descriptively because phase order, novelty, and temporal changes prevent causal attribution.

The target is 20 completers; 22–24 participants may be recruited to allow for attrition. Feasibility targets are at least 500 complete binary-labeled Phase 1 sessions overall, at least 20 labeled Phase 1 sessions per participant where browsing frequency permits, and at least 100 randomized elevated-risk Phase 2 sessions. These are operational targets rather than a formal power calculation. If recruitment, labels, or eligible intervention sessions fall substantially below target, the study will be framed as feasibility work and intervention estimates will be treated as exploratory.

5.2 Participants and Recruitment

Eligible participants are adults aged 18 years or older who use Chrome or a Chromium-based browser on a personal computer, regularly visit at least two domains they are willing to monitor, and can install a research extension for the study period. Recruitment may use a university convenience sample of students and knowledge workers. Participants are not recruited or screened on the basis of addiction, ADHD, mental health status, or other clinical categories. The paper will report recruitment source, demographics relevant to interpretation, attrition, usable-session counts, and missing-label rates without claiming representativeness.

5.3 Consent, Privacy, and Data Handling

Before installation, participants receive a consent form and field-level explanation of the extension. Consent covers configured domain hostnames, declared intention, intended duration, session timing, aggregate click, scroll, and keyboard-activity counts without key values, idle time, focus loss, activity-window summaries, post-session answers, and—during Phase 2—model version, risk estimate, random assignment, prompt display, and prompt response. The extension does not collect page text, passwords, screenshots, private messages, full browsing history, source code, keystroke values, webcam data, face identity, or emotion data.

Participants receive anonymous codes such as `P01`. They can pause monitoring, change monitored domains, inspect records, export their data, delete individual sessions, or delete all local data. Participant exports are stored in a private, access-controlled study directory outside the source repository. Only consented research staff can access the combined dataset. Any publication artifact will contain aggregates or de-identified excerpts rather than raw participant files. Ethics approval or institutional exemption, as applicable, must be obtained before recruitment.

5.4 Phase 1: Ten-Day Collection

After onboarding, each participant chooses monitored domains from both work/learning and mixed-use contexts. On every monitored-domain session, DriftSense asks for one of the seven neutral intention categories and optional intended duration. The participant then browses normally while aggregate session and 10-second activity-window signals are stored locally. At the session boundary, the extension asks whether the visit matched the declared intention. No mid-session risk estimate or reflective intervention is shown during Phase 1.

Participants use the collector for 10 consecutive study days. The research team checks only adherence and technical failures during collection; it does not inspect page content because none is collected. At the end of Phase 1, each participant exports the primary CSV and activity-window CSV. The exports are transferred using the consented study procedure, validated, and combined outside the repository.

5.5 Model Development and Freeze

Phase 1 days 1–7 form the development period and days 8–10 form a chronological holdout for known-user future-session performance. Model preprocessing, feature definitions, hyperparameters, and threshold rules are selected using days 1–7 only, with participant-grouped folds inside that development set. The untouched days 8–10 are evaluated once. A participant-grouped or leave-one-participant-out analysis over Phase 1 is reported separately as an estimate of performance for unseen users; it is not conflated with the same-participant chronological result.

The primary deployed candidate is intention-plus-activity logistic regression. It is compared with a time threshold, domain rule, intention-only logistic regression, and activity-only logistic regression; Random Forest is an optional offline robustness model. Prediction metrics are accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. Performance is computed at one, three, and five minutes and for the full session, but the intervention uses only the three-minute specification. Any session feature used at a cutoff is derived only from data observed by that time.

After holdout evaluation, one three-minute model, its preprocessing parameters, probability threshold, and maximum prompt rate are frozen. The threshold is chosen from development data to balance sensitivity against false prompts and expected burden; it is not tuned using Phase 2 outcomes. If the model performs near chance or produces an impractical eligible-session rate, the intervention may still be run as a technical feasibility pilot, but the paper will not claim successful prediction.

5.6 Phase 2: Seven-Day Micro-Randomized Intervention

The same participants use the intervention build for seven study days. Intention collection, passive signals, session boundaries, and post-session reflection remain identical to Phase 1. A session becomes randomization-eligible only when it remains active for three minutes, has the features required by the frozen model, exceeds the frozen threshold, and has not been suppressed by the prespecified prompt cap. The extension uses local random assignment with equal probability:

(1) **Reflective-prompt assignment.** Show *Still here for your original reason?* with *Continue intentionally* and *Finish now* actions.
(2) **Silent-control assignment.** Show no mid-session interface.

Assignment is logged before the prompt is rendered. Analysis follows assignment regardless of whether the participant interacts with the prompt. A maximum of one reflective prompt per session and a prespecified daily cap of three displayed prompts limit burden. Sessions below threshold, ending before three minutes, missing required features, or suppressed after the cap are retained for descriptive reporting but excluded from the randomized prompt-effect comparison. Both randomized assignments receive the same post-session reflection, preventing differences in label wording from defining the outcome.

5.7 Measures

The primary prediction outcomes are F1-score and ROC-AUC for intention-plus-activity prediction relative to time and domain baselines. Accuracy, precision, recall, confusion matrices, class balance, calibration where feasible, and the expected intervention rate at the frozen threshold are also reported. Results are shown separately for chronological same-participant prediction and participant-held-out prediction.

The primary intervention outcome is the proportion of complete binary-labeled eligible sessions reported as drift, compared by randomized assignment. Secondary behavioral outcomes are total session duration, duration after the three-minute assignment point, immediate session ending after a prompt, prompt response, prompt frequency, and post-session label completion. Phase 1 versus Phase 2 changes in drift rate and duration are descriptive secondary results only.

At the end of Phase 2, participants rate five statements on a five-point agreement scale: the prompts appeared at appropriate times; the prompts helped them reconsider their intention; the prompts were annoying; the extension was easy to use; and they would continue using it. One open-ended question asks what should change about prompt wording or timing. These measures assess acceptability and burden, not clinical benefit or objective attention.

5.8 Data Processing and Quality Checks

The preprocessing pipeline validates the exact participant CSV schema, confirms that each filename matches its participant identifier, rejects duplicate session identifiers and overlapping sessions, and reports incomplete or non-binary outcomes. Original participant exports are never modified. Binary analyses include only *Yes, it matched* (0) and *No, I drifted* (1); missing, *Continue intentionally*, and *Save for later* outcomes remain separate.

Session exports are joined to activity windows and intervention logs by `session_id`. Categorical features are one-hot encoded, and numeric features are standardized within training data for logistic regression. All preprocessing parameters are fitted on training partitions and applied unchanged to validation or test partitions. Quality reporting includes recruited and completed participants, sessions per participant, class distribution, label completion, files rejected, windows missing, intervention eligibility, assignment balance, prompt suppression, and attrition.

5.9 Analysis Plan

For RQ1, model performance is summarized across the predefined splits with 95% confidence intervals obtained by resampling participants where feasible. The main comparison asks whether intention-plus-activity logistic regression improves F1-score and ROC-AUC over time and domain baselines. Intention-only and activity-only models show the incremental contribution of each feature group. A result is not interpreted as generalization to new users unless participant-held-out performance supports that claim.

For RQ2, analysis is restricted to randomized eligible sessions with binary post-session labels and follows assigned condition. For each participant with observations in both assignments, drift proportions are calculated separately for reflective-prompt and silent-control sessions. The prespecified simple comparison is a paired Wilcoxon signed-rank test on participant-level drift proportions, accompanied by the paired difference and a 95% participant-bootstrap confidence interval. Assignment-specific missing-label rates are reported. If the number of eligible sessions supports it, a mixed-effects logistic regression with a participant random intercept and prompt assignment as the main fixed effect is added as a sensitivity analysis.

Secondary outcomes are analyzed descriptively or with paired participant-level comparisons. Open-ended responses are summarized with a transparent lightweight content analysis. No individual mechanism is declared effective from a Phase 1–Phase 2 before–after difference alone. Results will be described as short-term and preliminary; a seven-day intervention cannot establish durable behavior change. No results, significance values, or effect sizes will be inserted until the real consented dataset has been analyzed.

5.10 Reproducibility and Reporting Commitments

Before recruitment, the study materials will specify the session boundary rules, seven intention options, reflection wording, binary-label mapping, prompt cap, exclusion rules, and the distinction between confirmatory and exploratory outcomes. Before Phase 2, the extension build will record a versioned model artifact, preprocessing specification, decision threshold, and randomization procedure. The analysis will produce a participant-flow summary and an accountable session-flow summary from all recorded sessions to the final RQ1 and RQ2 analysis sets. Counts will be reported for sessions ending before three minutes, missing reflection labels, incomplete activity windows, threshold eligibility, suppression by the daily cap, randomized assignment, and prompt delivery failures.

Code, schemas, isolated non-participant test fixtures, and analysis scripts may be released after review, but raw participant exports will not be placed in the public repository. A de-identified derived dataset will be shared only if consent, ethics approval, and disclosure-risk review permit it. Otherwise, reproducibility will rely on field-level documentation, deterministic preprocessing, fixed random seeds, environment information, and aggregate or disclosure-safe outputs.

5.11 Anticipated Validity Boundaries

The primary construct is a participant's post-session judgment of alignment, not an objective measurement of attention. Labels may vary across people and may be influenced by recall, social desirability, repeated prompting, or the intervention itself. Reporting label-completion and non-binary outcomes by assignment is therefore part of the main validity assessment rather than a housekeeping detail.

The causal estimate for RQ2 applies only to sessions that meet the frozen model's eligibility rule and are not suppressed by the prompt cap. It does not estimate the effect of prompting every browser session or every user. Randomization protects the prompt-versus-silent comparison within this eligible set, while the fixed phase order prevents a causal interpretation of the overall Phase 1–Phase 2 change. A 15–20-person convenience sample also limits precision and generalization, particularly for participant-held-out prediction. Finally, a seven-day intervention can establish short-term feasibility and preliminary efficacy only; it cannot demonstrate persistence after the extension is removed. These boundaries will remain explicit even if the observed effects are favorable.

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